

Interactions between histone subunits (which are positively charged at physiological pH) and negatively charged nucleic acids are altered by acetylation and methylation of histone subunits. These different post-translational modifications are recognized by histone-binding proteins such as BROMO domain Adjacent to Zinc finger 2B (BAZ2B). BAZ2B has a net negative charge under physiological conditions but contains a lysine-rich regulatory region between its two histone-binding domains.

NMR data revealed potential interactions between BAZ2B and the histone subunit H3 (Figure 1). Binding was confirmed by isothermal titration calorimetry (ITC). ITC measures enthalpy of binding when a protein solution is titrated with ligand injections of equal volume and concentration; each injection results in a negative peak as heat is released. Figure 2 shows ITC results from an experiment in which purified wild-type (WT) BAZ2B or BAZ2B with the regulatory region removed (10M+) was titrated with unmodified H3 or with H3 that was acetylated at lysine-14 (H3K14ac). In each case, the Hill coefficient was approximately 1.

To test binding in a biological setting, lysates from cells that expressed either WT or 10M+ BAZ2B were run on affinity columns containing either H3 or H3K14ac covalently attached to beads. The column was washed several times, and then bound BAZ2B was eluted and quantified by western blot analysis (Figure 3).

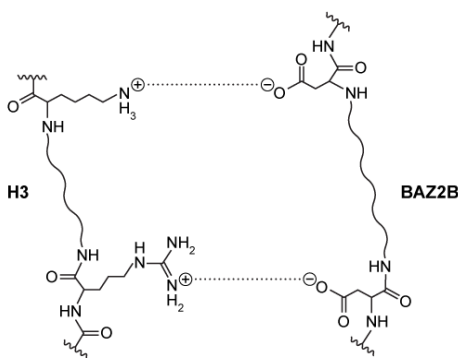


Figure 1 Electrostatic interactions between H3 and BAZ2B amino acid residues

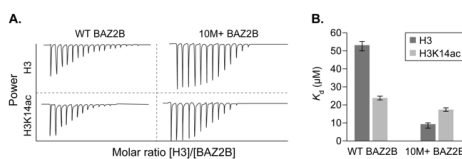


Figure 2 ITC traces (A) and K_d values (B) of WT BAZ2B and 10M+ BAZ2B titrated with either unmodified H3 or H3K14ac

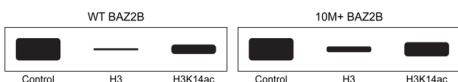


Figure 3 Western blots of WT and 10M+ BAZ2B recovered from columns containing unmodified histone subunit 3 (the lanes labeled H3) or acetylated H3 (the lanes labeled H3K14ac). Control lanes contain purified BAZ2B to show that antibody binding was successful.

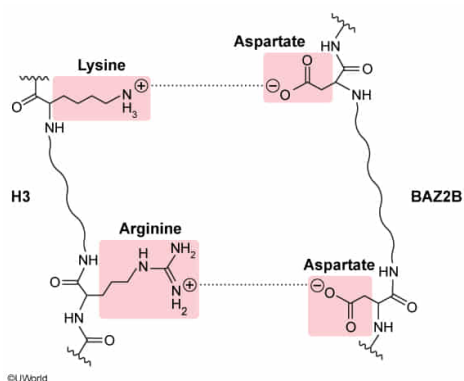
Based on information in the passage, NMR data indicated electrostatic interactions involving all of the following amino acid residues EXCEPT:

- ☐ A. lysine. [7%]
- ☐ B. aspartate. [13%]
- ☐ C. arginine. [16%]
- ☒ D. glutamate. [62%]

Correct

63% answered correctly

Explanation:



All amino acids have a **common backbone** structure, but each has a **unique side chain** that determines its identity. The **positively charged side chains** of basic amino acids can **interact electrostatically** (ionic bonds) with the **negatively charged side chains** of acidic amino acids. These interactions can stabilize the **tertiary structure** of a protein as well as the interactions between proteins.

The passage states that NMR revealed potential electrostatic interactions between BAZ2B and H3, and the interactions are shown in Figure 1. This illustration displays the four amino acid residues involved in the interaction between BAZ2B and H3 as well as some neighboring backbone atoms. The identities of the four amino acid residues can be determined by focusing on the side chain atoms: the two positively charged amino acids in H3, **lysine** and **arginine (Choices A and C)**, interacted with the two negatively charged **aspartate** side chains in BAZ2B (**Choice B**).

The question asks for the residue that is NOT depicted. **Glutamate** differs from aspartate in that it has an **additional methylene (CH₂)** unit before the carboxyl group in its side chain. Although glutamate is negatively charged and could potentially interact electrostatically with arginine or lysine, the data obtained from NMR did not show any glutamate residues involved in H3-BAZ2B interactions.

Educational objective:

Amino acids can be identified by their unique side chain structures. Positively charged amino acids such as lysine and arginine can interact with negatively charged amino acids such as aspartate and glutamate.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402196
System:
Amino Acids and Proteins

Interactions between histone subunits (which are positively charged at physiological pH) and negatively charged nucleic acids are altered by acetylation and methylation of histone subunits. These different post-translational modifications are recognized by histone-binding proteins such as BROMO domain Adjacent to Zinc finger 2B (BAZ2B). BAZ2B has a net negative charge under physiological conditions but contains a lysine-rich regulatory region between its two histone-binding domains.

NMR data revealed potential interactions between BAZ2B and the histone subunit H3 (Figure 1). Binding was confirmed by isothermal titration calorimetry (ITC). ITC measures enthalpy of binding when a protein solution is titrated with ligand injections of equal volume and concentration; each injection results in a negative peak as heat is released. Figure 2 shows ITC results from an experiment in which purified wild-type (WT) BAZ2B or BAZ2B with the regulatory region removed (10M+) was titrated with unmodified H3 or with H3 that was acetylated at lysine-14 (H3K14ac). In each case, the Hill coefficient was approximately 1.

To test binding in a biological setting, lysates from cells that expressed either WT or 10M+ BAZ2B were run on affinity columns containing either H3 or H3K14ac covalently attached to beads. The column was washed several times, and then bound BAZ2B was eluted and quantified by western blot analysis (Figure 3).

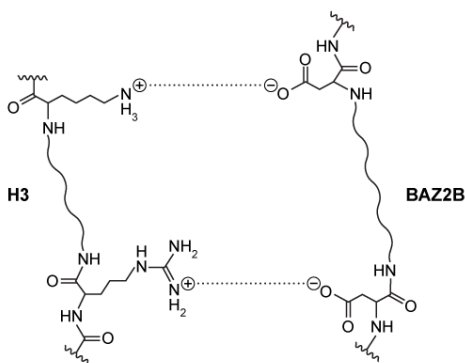


Figure 1 Electrostatic interactions between H3 and BAZ2B amino acid residues

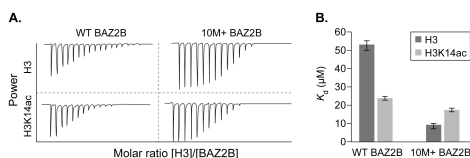


Figure 2 ITC traces (A) and K_d values (B) of WT BAZ2B and 10M+ BAZ2B titrated with either unmodified H3 or H3K14ac

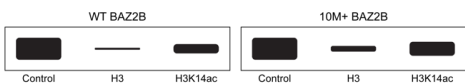


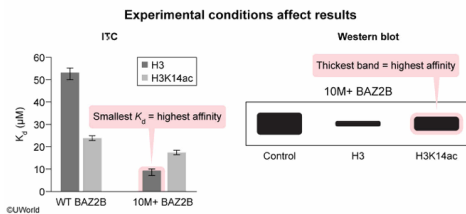
Figure 3 Western blots of WT and 10M+ BAZ2B recovered from columns containing unmodified histone subunit 3 (the lanes labeled H3) or acetylated H3 (the lanes labeled H3K14ac). Control lanes contain purified BAZ2B to show that antibody binding was successful.

Based on Figures 2 and 3, which form of the histone subunit has the highest affinity for 10M+ BAZ2B as revealed by the ITC and western blot results?

- ☒ A. H3 in the ITC experiment and H3K14ac in the western blot analysis [57%]
- ☐ B. H3K14ac in the ITC experiment and H3 in the western blot analysis [5%]
- ☐ C. H3K14ac in both ITC and western blot [28%]
- ☐ D. H3 in both ITC and western blot [9%]

Correct

57% answered correctly

Explanation:

Biochemical assays commonly measure the parameters of purified biomolecules such as proteins. They are tightly controlled experiments that allow variables to be isolated because only a few types of molecule are present. However, these assays may not accurately reflect the behavior of a complex biological system, in which thousands of different molecules are present. The passage illustrates a case in which results from two such experiments do not fully agree.

In the ITC experiment, the dissociation constant K_d was measured and used to determine the binding affinity of purified BAZ2B for different forms of the histone subunit H3. The **low K_d** of the 10M+ BAZ2B interaction with H3 shows that it has a **higher binding affinity for H3** than for H3K14ac, as measured by ITC.

In the second experiment, proteins were extracted directly from cells (ie, they were not purified). When the **affinity column** contained unmodified H3, a small band was produced on the western blot. This suggests that a significant amount of 10M+ BAZ2B *failed* to bind to the column and washed out prior to the elution step, which was used for the western blot. Therefore, a small band indicates low affinity for the column. In contrast, when the column contained H3K14ac, a thicker band was produced, indicating that 10M+ BAZ2B bound tightly to the column and was *not* washed out prior to elution for the western blot. Therefore, 10M+ BAZ2B obtained from cell lysates has **higher affinity for H3K14ac than for H3**.

(Choice B) This option is the opposite of the correct answer. 10M+ BAZ2B binds H3 with the highest affinity in the ITC experiment but H3K14ac in the western blot analysis.

(Choices C and D) 10M+ BAZ2B shows *different* relative affinities for H3 and H3K14ac in each experiment.

Educational objective:

Assays with purified proteins are useful because they can isolate variables, but they may not always reflect protein behavior in a biological setting. Binding affinity can be measured by the dissociation constant K_d ,

where a low K_d corresponds to high binding affinity. It can also be assessed qualitatively by affinity chromatography and western blot analysis, in which a thick, dense band indicates more protein and therefore higher affinity.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402197
System:
Amino Acids and Proteins

Interactions between histone subunits (which are positively charged at physiological pH) and negatively charged nucleic acids are altered by acetylation and methylation of histone subunits. These different post-translational modifications are recognized by histone-binding proteins such as BROMO domain Adjacent to Zinc finger 2B (BAZ2B). BAZ2B has a net negative charge under physiological conditions but contains a lysine-rich regulatory region between its two histone-binding domains.

NMR data revealed potential interactions between BAZ2B and the histone subunit H3 (Figure 1). Binding was confirmed by isothermal titration calorimetry (ITC). ITC measures enthalpy of binding when a protein solution is titrated with ligand injections of equal volume and concentration; each injection results in a negative peak as heat is released. Figure 2 shows ITC results from an experiment in which purified wild-type (WT) BAZ2B or BAZ2B with the regulatory region removed (10M+) was titrated with unmodified H3 or with H3 that was acetylated at lysine-14 (H3K14ac). In each case, the Hill coefficient was approximately 1.

To test binding in a biological setting, lysates from cells that expressed either WT or 10M+ BAZ2B were run on affinity columns containing either H3 or H3K14ac covalently attached to beads. The column was washed several times, and then bound BAZ2B was eluted and quantified by western blot analysis (Figure 3).

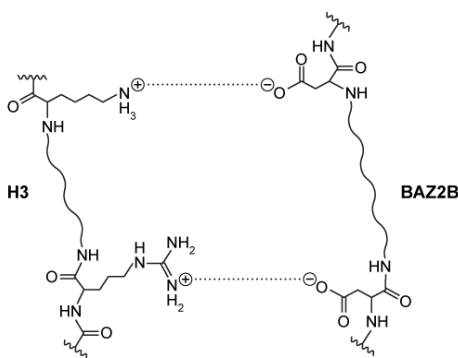


Figure 1 Electrostatic interactions between H3 and BAZ2B amino acid residues

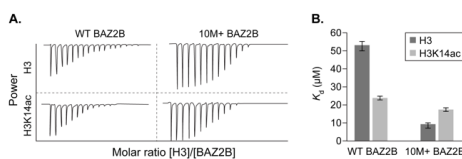


Figure 2 ITC traces (A) and K_d values (B) of WT BAZ2B and 10M+ BAZ2B titrated with either unmodified H3 or H3K14ac

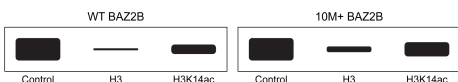


Figure 3 Western blots of WT and 10M+ BAZ2B recovered from columns containing unmodified histone subunit 3 (the lanes labeled H3) or acetylated H3 (the lanes labeled H3K14ac). Control lanes contain purified BAZ2B to show that antibody binding was successful.

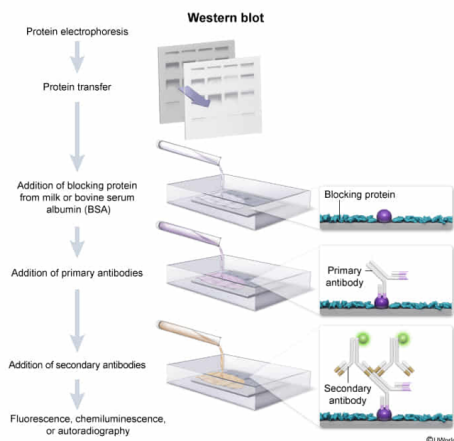
Which step is NOT required to generate the western blot shown in Figure 3?

- ☐ A. Gel electrophoresis of BAZ2B proteins eluted from affinity columns [19%]
- ☐ B. Transfer of proteins from a gel to a protein-binding membrane [11%]
- ☐ C. Blocking nonspecific binding between antibodies and the blotting membrane [31%]
- ☒ D. Incubation of the blotting membrane with antibodies against histone subunit H3 [37%]

Correct

37% answered correctly

Explanation:



Western blot analysis is a technique used to detect the presence of a specific protein and compare its relative abundance in one set of conditions to that obtained under other conditions. It is performed in the following steps:

1. Samples containing the protein of interest are loaded onto a gel and subjected to **electrophoresis**, causing the protein to migrate through the gel (**Choice A**).
2. Proteins in the gel are **transferred** to a **protein-binding membrane** such as nitrocellulose or polyvinylidene fluoride (PVDF), where they become immobilized (**Choice B**). Transfer is usually carried out by electrophoresis.
3. The portions of the membrane to which protein was *not* transferred are **blocked** by protein-rich mixtures such as bovine serum albumin (BSA) or low-fat milk. This step prevents antibodies, which are themselves proteins, from **nonspecifically binding** to the nitrocellulose in the next step (**Choice C**).
4. The membrane is incubated with **antibodies** that specifically **bind the protein of interest**. These antibodies may be labeled with a marker and detected directly, but more commonly they are subjected to a labeled secondary antibody that specifically binds the first (primary) antibody. Use of secondary antibodies helps **improve detection**.
5. Antibodies are **detected** by techniques such as **fluorescence** or **chemiluminescence**.

In this case, the **protein of interest is BAZ2B**, which was eluted from the **affinity column**. According to the passage, H3 proteins were covalently linked to the beads in the column and therefore they do not elute. Instead, the purpose of H3 (or H3K14ac) was to bind as much BAZ2B to the column as possible *prior* to elution. As such, the **primary antibodies** must be **specific to BAZ2B, not H3**, to detect the amount of BAZ2B that was bound by the affinity column.

Educational objective:

Western blot analysis is a technique used to detect the presence of specific proteins. It is performed by gel electrophoresis of the protein of interest, followed by transfer to a protein-binding membrane (eg, nitrocellulose) and incubation with antibodies specific to the protein of interest. The primary antibody or a labeled secondary antibody is then detected by fluorescence or chemiluminescence.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402198
System:
Amino Acids and Proteins

Interactions between histone subunits (which are positively charged at physiological pH) and negatively charged nucleic acids are altered by acetylation and methylation of histone subunits. These different post-translational modifications are recognized by histone-binding proteins such as BROMO domain Adjacent to Zinc finger 2B (BAZ2B). BAZ2B has a net negative charge under physiological conditions but contains a lysine-rich regulatory region between its two histone-binding domains.

NMR data revealed potential interactions between BAZ2B and the histone subunit H3 (Figure 1). Binding was confirmed by isothermal titration calorimetry (ITC). ITC measures enthalpy of binding when a protein solution is titrated with ligand injections of equal volume and concentration; each injection results in a negative peak as heat is released. Figure 2 shows ITC results from an experiment in which purified wild-type (WT) BAZ2B or BAZ2B with the regulatory region removed (10M+) was titrated with unmodified H3 or with H3 that was acetylated at lysine-14 (H3K14ac). In each case, the Hill coefficient was approximately 1.

To test binding in a biological setting, lysates from cells that expressed either WT or 10M+ BAZ2B were run on affinity columns containing either H3 or H3K14ac covalently attached to beads. The column was washed several times, and then bound BAZ2B was eluted and quantified by western blot analysis (Figure 3).

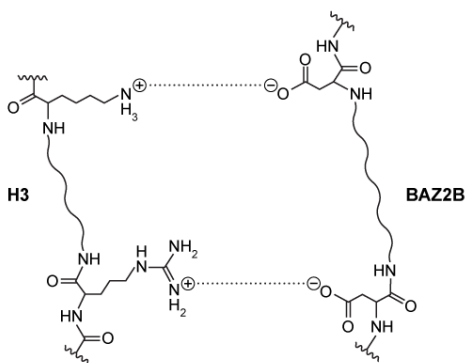


Figure 1 Electrostatic interactions between H3 and BAZ2B amino acid residues

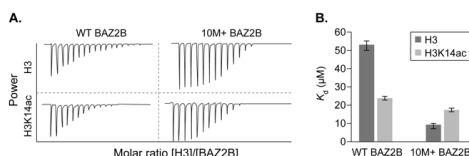


Figure 2 ITC traces (A) and K_d values (B) of WT BAZ2B and 10M+ BAZ2B titrated with either unmodified H3 or H3K14ac

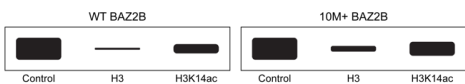


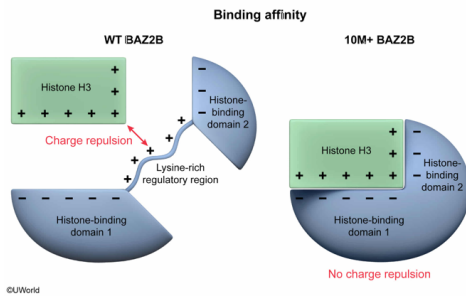
Figure 3 Western blots of WT and 10M+ BAZ2B recovered from columns containing unmodified histone subunit 3 (the lanes labeled H3) or acetylated H3 (the lanes labeled H3K14ac). Control lanes contain purified BAZ2B to show that antibody binding was successful.

Which statement best explains why the 10M+ form of BAZ2B binds H3 with greater affinity than its wild-type counterpart?

- ☐ A. The increased hydrophobicity of 10M+ BAZ2B induces stronger London forces with H3. [7%]
- ☒ B. The loss of positive charges on 10M+ BAZ2B improves its electrostatic interactions with H3. [70%]
- ☐ C. 10M+ BAZ2B is more readily acetylated than WT BAZ2B. [17%]
- ☐ D. WT BAZ2B is expressed at lower levels than 10M+ BAZ2B *in vivo*. [4%]

Correct

74% answered correctly

Explanation:

Proteins predominantly **interact with each other** through **noncovalent interactions**. An **increased number** of these interactions or a decrease in the number of repulsive interactions leads to **increased overall strength** of binding.

According to the passage, histones are positively charged and BAZ2B is negatively charged, which allows them to interact electrostatically. However, BAZ2B also contains a lysine-rich regulatory region that links the two histone-binding domains. The lysine residues in this region contribute positive charge, which weakens interactions with positively charged histones. In 10M+ BAZ2B, the regulatory region is removed. The resulting **absence of several positive charges** from lysine residues yields a more negative net charge on BAZ2B, **improving its electrostatic interactions** with the positive charges of H3.

(Choice A) **London forces** arise when the electron clouds that surround atoms become distorted, giving a partial positive charge on one side of the atom and a partial negative charge on the other. Although removing lysine residues results in a local increase in hydrophobicity near the linker region, it also decreases the number of electron clouds available and therefore *reduces* the number of London forces.

(Choice C) The passage describes acetylation of H3 but does not indicate that BAZ2B is ever acetylated.

(Choice D) The **control** lanes of the two western blots indicate that WT and 10M+ BAZ2B are expressed in cells at approximately equal levels. In addition, expression level does not alter affinity.

Educational objective:

Proteins can interact with each other, largely through noncovalent forces such as ionic interactions. These interactions can be strengthened by increasing the number of attractive forces or by decreasing the number of repulsive forces.

Time spent: 0 sec

Subject:
Biochemistry

QID: 402199

System:
Amino Acids and Proteins

Interactions between histone subunits (which are positively charged at physiological pH) and negatively charged nucleic acids are altered by acetylation and methylation of histone subunits. These different post-translational modifications are recognized by histone-binding proteins such as BROMO domain Adjacent to Zinc finger 2B (BAZ2B). BAZ2B has a net negative charge under physiological conditions but contains a lysine-rich regulatory region between its two histone-binding domains.

NMR data revealed potential interactions between BAZ2B and the histone subunit H3 (Figure 1). Binding was confirmed by isothermal titration calorimetry (ITC). ITC measures enthalpy of binding when a protein solution is titrated with ligand injections of equal volume and concentration; each injection results in a negative peak as heat is released. Figure 2 shows ITC results from an experiment in which purified wild-type (WT) BAZ2B or BAZ2B with the regulatory region removed (10M+) was titrated with unmodified H3 or with H3 that was acetylated at lysine-14 (H3K14ac). In each case, the Hill coefficient was approximately 1.

To test binding in a biological setting, lysates from cells that expressed either WT or 10M+ BAZ2B were run on affinity columns containing either H3 or H3K14ac covalently attached to beads. The column was washed several times, and then bound BAZ2B was eluted and quantified by western blot analysis (Figure 3).

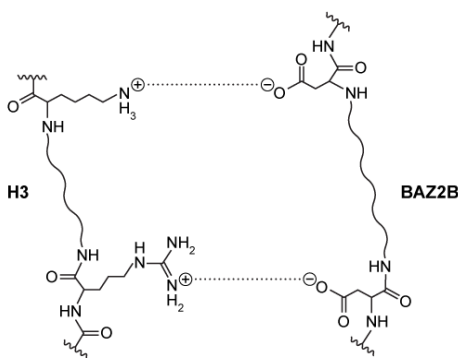


Figure 1 Electrostatic interactions between H3 and BAZ2B amino acid residues

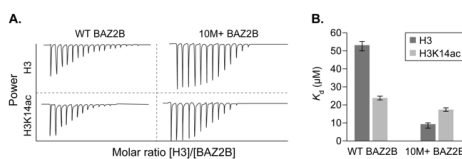


Figure 2 ITC traces (A) and K_d values (B) of WT BAZ2B and 10M+ BAZ2B titrated with either unmodified H3 or H3K14ac

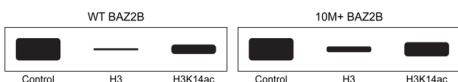


Figure 3 Western blots of WT and 10M+ BAZ2B recovered from columns containing unmodified histone subunit 3 (the lanes labeled H3) or acetylated H3 (the lanes labeled H3K14ac). Control lanes contain purified BAZ2B to show that antibody binding was successful.

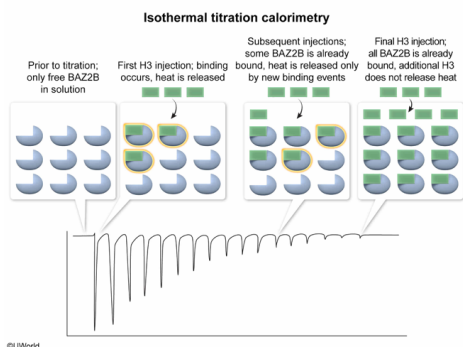
Based on the information in the passage, why does the size of the ITC peaks decrease over the course of each experiment in Figure 2?

- ☐ A. Increasing H3 concentration pushes the equilibrium toward dissociation. [12%]
- ☒ B. Fewer binding events occur on BAZ2B as the experiment progresses. [60%]
- ☐ C. Less H3 is present in each subsequent injection as the titration progresses. [19%]
- ☐ D. Negative cooperativity exhibited by BAZ2B causes binding to decrease as H3 is added. [8%]

Correct

59% answered correctly

Explanation:



According to the passage, isothermal titration calorimetry (ITC) measures the enthalpy (heat) of binding as increasing concentrations of a ligand are titrated into a solution of protein. Each **binding event releases heat (negative ΔH)** and is displayed as a negative peak on the ITC trace.

Each H3 injection results in some amount of binding and heat release and leaves **fewer available binding sites** for the next injection. More BAZ2B binding sites are filled after each injection, so the probability of new binding events in subsequent injections decreases. Therefore, **less heat is released** per injection as BAZ2B becomes **saturated**. This effect is reflected as a progressive decrease in the size of the peaks. Once all binding sites are filled, additional H3 injections cannot induce any more binding, so no more heat is released and the peaks disappear.

(Choice A) Based on [Le Châtelier's principle](#), as more reactant (H3) is added, the equilibrium shifts toward product *formation* (binding), not dissociation.

(Choice C) The passage states that each injection is of equal volume and concentration, so the amount of H3 in each injection is constant.

(Choice D) Cooperativity, described by the [Hill coefficient](#), refers to the ability of ligand binding to alter a protein's affinity for additional ligands. The passage states that the Hill coefficient for each titration was approximately 1, indicating no cooperativity.

Educational objective:

Binding interactions release heat (negative ΔH). As a protein becomes saturated with ligand, additional ligand is unable to induce more binding, so the magnitude of the heat released decreases as more ligand is added.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402200
System:
Amino Acids and Proteins

Interactions between histone subunits (which are positively charged at physiological pH) and negatively charged nucleic acids are altered by acetylation and methylation of histone subunits. These different post-translational modifications are recognized by histone-binding proteins such as BROMO domain Adjacent to Zinc finger 2B (BAZ2B). BAZ2B has a net negative charge under physiological conditions but contains a lysine-rich regulatory region between its two histone-binding domains.

NMR data revealed potential interactions between BAZ2B and the histone subunit H3 (Figure 1). Binding was confirmed by isothermal titration calorimetry (ITC). ITC measures enthalpy of binding when a protein solution is titrated with ligand injections of equal volume and concentration; each injection results in a negative peak as heat is released. Figure 2 shows ITC results from an experiment in which purified wild-type (WT) BAZ2B or BAZ2B with the regulatory region removed (10M+) was titrated with unmodified H3 or with H3 that was acetylated at lysine-14 (H3K14ac). In each case, the Hill coefficient was approximately 1.

To test binding in a biological setting, lysates from cells that expressed either WT or 10M+ BAZ2B were run on affinity columns containing either H3 or H3K14ac covalently attached to beads. The column was washed several times, and then bound BAZ2B was eluted and quantified by western blot analysis (Figure 3).

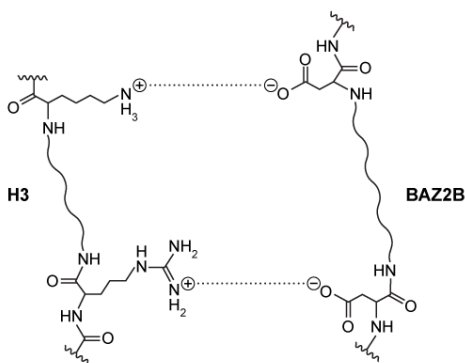


Figure 1 Electrostatic interactions between H3 and BAZ2B amino acid residues

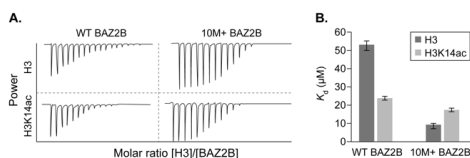


Figure 2 ITC traces (A) and K_d values (B) of WT BAZ2B and 10M+ BAZ2B titrated with either unmodified H3 or H3K14ac

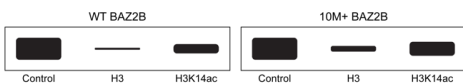


Figure 3 Western blots of WT and 10M+ BAZ2B recovered from columns containing unmodified histone subunit 3 (the lanes labeled H3) or acetylated H3 (the lanes labeled H3K14ac). Control lanes contain purified BAZ2B to show that antibody binding was successful.

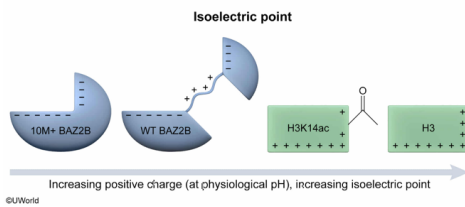
Which of the proteins mentioned in the passage has the highest isoelectric point?

- ☐ A. WT BAZ2B [15%]
- ☐ B. 10M+ BAZ2B [15%]
- ☒ C. H3 [53%]
- ☐ D. H3K14ac [16%]

Correct

52% answered correctly

Explanation:



The **isoelectric point (pI)** of a protein is the pH at which it has a **net charge of zero**. When the pI is reached, individual residues may still be positively or negatively charged, but there must be an equal number of positive and negative charges so they cancel each other. Proteins with a **net negative charge** at physiological pH have a **low pI** and require a low pH for neutralization, whereas a protein with a **net positive charge** at physiological pH has a **high pI** because a high pH is needed to deprotonate enough of the positively charged side chains.

According to the passage, BAZ2B has a net negative charge under physiological conditions. In contrast, the first sentence of the passage describes histone subunits as positively charged at physiological pH. Therefore, both forms of H3 have a higher pI than either form of BAZ2B. The passage also indicates that H3K14ac is acetylated at a lysine residue. **Histone acetylation** neutralizes the positive charge of lysine, so H3K14ac has one fewer positive charge than unmodified H3. The pI of H3K14ac is decreased relative to the pI of unmodified H3, so **H3 has the highest pI** of the proteins listed.

(Choices A and B) WT BAZ2B and 10M+ BAZ2B have net negative charges at physiological pH. Therefore, both have a lower pI than the H3 subunits, which are positively charged at physiological pH.

(Choice D) H3K14ac is less positively charged than unmodified H3 at physiological pH, so it does not have the highest pI.

Educational objective:

The isoelectric point (pI) of a protein is the pH at which the protein has a net charge of zero. The more positively charged the protein is at physiological pH, the higher its pI.

Time spent: 0 sec

Subject:
Biochemistry

QID: 402201

System:
Amino Acids and Proteins